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Studierichting: Informatica
Oefeningenreeks 4A nr. 15.5

$$\int (x^2 - x - 4)e^x dx$$

$$\left(\int udv = uv - \int vdu\right)$$

$$\left\{\begin{array}{l} u = x^2 - x - 4 \\ dv = e^x dx \end{array}\right. \Rightarrow \left\{\begin{array}{l} du = (2x-1)dx \\ v = e^x \end{array}\right.$$

$$= (x^2 - x - 4)e^x - \int (2x - 2)e^x dx$$

$$\left(\int udv = uv - \int vdu\right)$$

$$\left\{\begin{array}{l} u = 2x - 1 \\ dv = e^d x \end{array}\right. \Rightarrow \left\{\begin{array}{l} du = 2dx \\ v = e^x \end{array}\right.$$

$$= (x^2 - x - 4)e^x - (2x - 1)e^x - \int 2e^x dx$$

$$= (x^2 - x - 4)e^x - (2x - 1)e^x - 2e^x$$

$$= (x^2 - x - 4)e^x - (2x - 3)e^x$$

$$= (x^2 - x - 4 - 2x + 3)e^x$$

$$= (x^2 - 3x - 1)e^x + C$$

References